Multiple Regression: Moderation and Mediation Tutorial

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Load in required libraries

```
library(jmv)
library(rockchalk)
library(psych)
library(multilevel)
library(tidyverse)
library(interactions)
```

Load data set into R-Environment

```
dat<-read.csv("DemoMod.csv")
dat2<-read.csv("DemoMed.csv")
```

Moderation Tutorial Markdown

We are going to use `dat` for a tutorial of moderation analyses. This data set consists of performance evaluation review data (*perf_eval*) for employees of a Marketing Firm. Additionally, we have data for the employee's original interview rating (*interview*), their age at hire (*age*), dichotomous coded sex (*gender*: 0 = Males; 1 = Females), and polytomous race (*race*: 1 = Caucasian, 2 = African-American, 3 = Asian).

Applied prompt: Our boss wants to know if the association between performance evaluation and interview ratings is moderated by the demographic variables of age and gender. We will treat performance evaluation as the criterion variable for this tutorial.

Recode Predictors and Moderators (Centering and Dummy Coding)

```
# Center all predictors/moderators that are continuous
dat$interviewC <- dat$interview - mean(dat$interview)
dat$ageC <- dat$age - mean(dat$age)

# Dummy code all categorical variables NOT already coded 0 and 1
# These will NOT be used in the analyses, but you could use them to expand your understanding
AA_dc <- dummy.code(dat$race, group = 2)
As_dc <- dummy.code(dat$race, group = 3)
dat <- data.frame(dat, AA_dc, As_dc)

# Create our interaction term(s)
dat$int_gender <- dat$interviewC*dat$gender
dat$int_age <- dat$interviewC*dat$ageC
```

Calculate Descriptive Statistics and Correlation Matrix

```
dat$genderF <- factor(dat$gender,
                      levels = c(0,1),
                      labels = c("Women","Men"))

describe(dat[2:4])
```

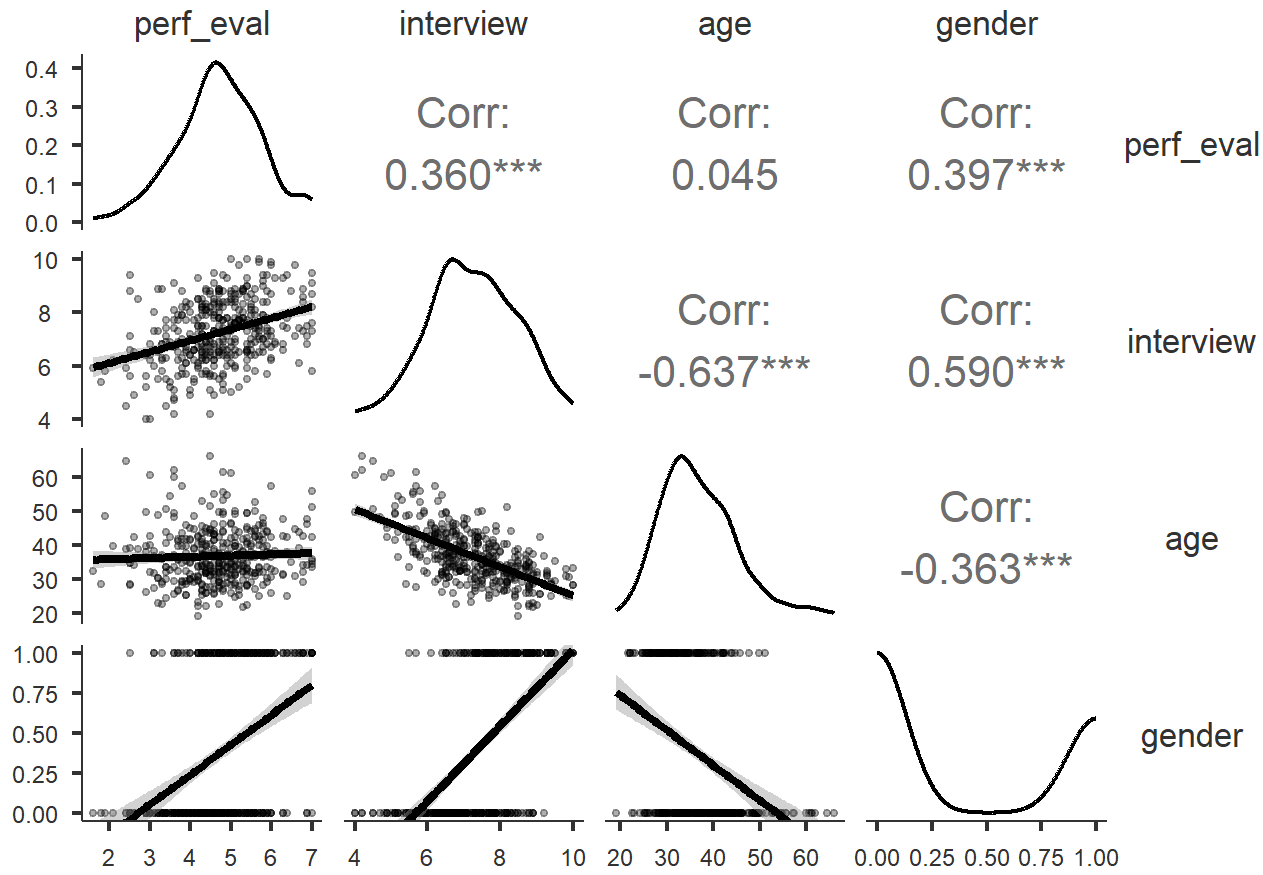
```
##           vars   n  mean   sd median trimmed  mad  min  max range  skew
## perf_eval    1 377  4.71 1.03    4.7    4.72 1.04  1.6  7.0   5.4 -0.13
## interview    2 377  7.26 1.21    7.3    7.26 1.33  4.0 10.0   6.0 -0.05
## age          3 377 36.74 8.03   35.8   36.16 7.71 19.1 66.1  47.0  0.77
##           kurtosis   se
## perf_eval     0.04 0.05
## interview    -0.37 0.06
## age           0.79 0.41
```

```
describeBy(dat[2:4],
           group = dat$genderF,
           mat=TRUE)
```

```
##            item group1 vars   n      mean        sd median   trimmed     mad
## perf_eval1    1  Women    1 237  4.391983 0.9622422   4.40  4.412565 0.88956
## perf_eval2    2    Men    1 140  5.235000 0.9136790   5.25  5.233036 0.81543
## interview1    3  Women    2 237  6.709283 1.0022846   6.70  6.717801 0.88956
## interview2    4    Men    2 140  8.179286 0.9258476   8.20  8.198214 0.88956
## age1          5  Women    3 237 38.983122 8.2834920  38.10 38.447644 7.85778
## age2          6    Men    3 140 32.954286 5.9081991  32.20 32.627679 6.00453
##             min  max range        skew    kurtosis         se
## perf_eval1  1.6  7.0   5.4 -0.17339606  0.09600617 0.06250435
## perf_eval2  2.5  7.0   4.5 -0.08408551 -0.09769623 0.07721997
## interview1  4.0  9.2   5.2 -0.08274133 -0.05116894 0.06510538
## interview2  5.5 10.0   4.5 -0.22775470 -0.23778385 0.07824840
## age1       19.1 66.1  47.0  0.64005520  0.53172073 0.53807062
## age2       21.6 51.1  29.5  0.53625173  0.09885299 0.49933396
```

```
corrMatrix(dat[,c(2:5)],
           plots = TRUE,
           plotDens = TRUE,
           plotStats = TRUE)$plot
```

```
## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
## ℹ Please use the `linewidth` argument instead.
## ℹ The deprecated feature was likely used in the jmvcore package.
##   Please report the issue at <https://github.com/jamovi/jmvcore/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Moderation | `linReg()` Technique Interview By Gender Moderation

```
linReg(data = dat,
       dep = 'perf_eval',
       covs = c('interviewC', 'gender','int_gender'),
       blocks = list(
         list('interviewC', 'gender'),
         list('int_gender')),
       ci = TRUE,
       modelTest = TRUE,
       r2Adj = TRUE,
       stdEst = TRUE,
       ciStdEst = TRUE)
```

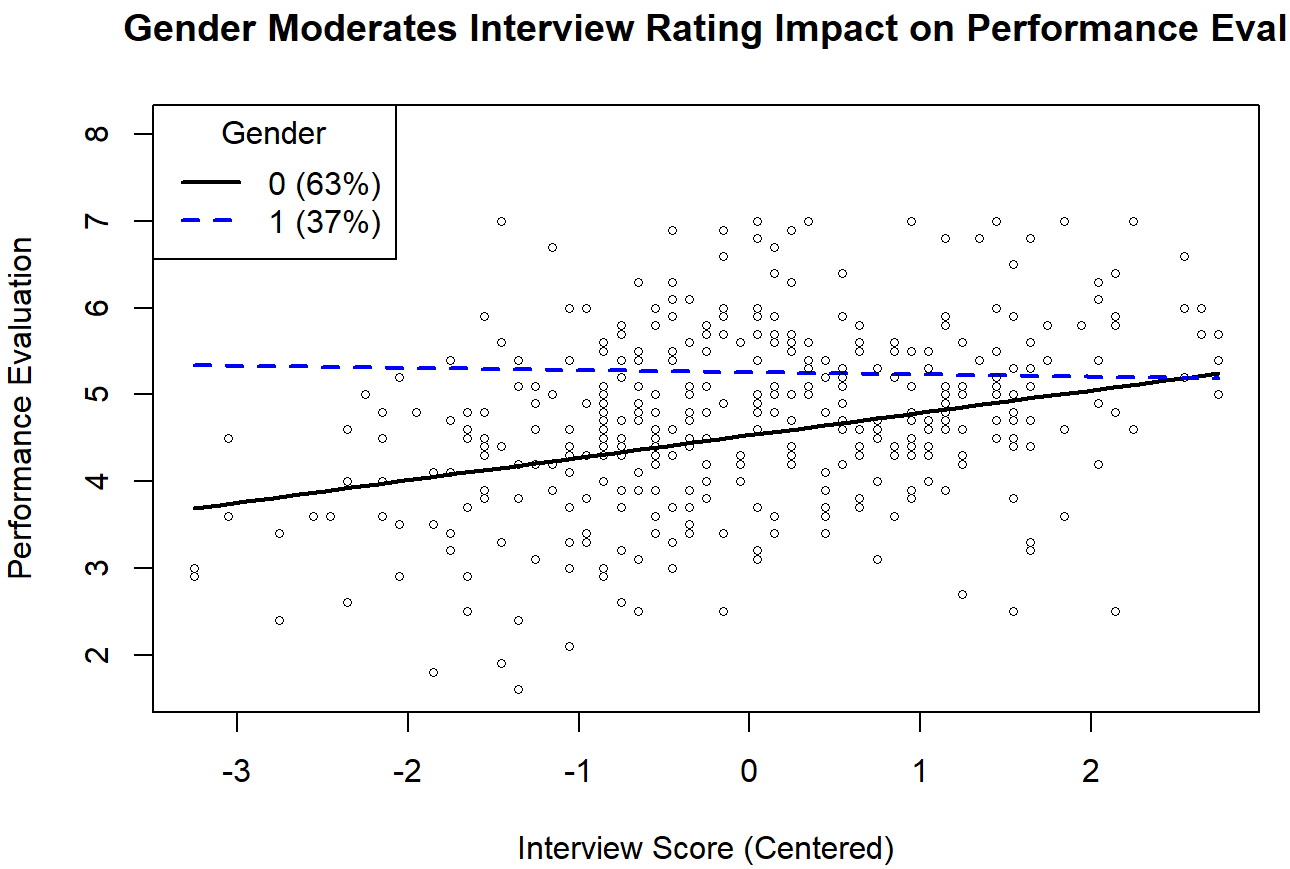
```
##
##  LINEAR REGRESSION
##
##  Model Fit Measures
##  ──────────────────────────────────────────────────────────────────────────────────────────
##    Model    R            R²           Adjusted R²    F           df1    df2    p
##  ──────────────────────────────────────────────────────────────────────────────────────────
##        1    0.4262549    0.1816932      0.1773173    41.52066      2    374    < .0000001
##        2    0.4447164    0.1977727      0.1913204    30.65183      3    373    < .0000001
##  ──────────────────────────────────────────────────────────────────────────────────────────
##    Note. Models estimated using sample size of N=377
##
##
##  Model Comparisons
##  ──────────────────────────────────────────────────────────────────────────────────────────
##    Model         Model    ΔR²           F           df1    df2    p
##  ──────────────────────────────────────────────────────────────────────────────────────────
##        1    -        2    0.01607944    7.476224      1    373    0.0065501
##  ──────────────────────────────────────────────────────────────────────────────────────────
##
##
##  MODEL SPECIFIC RESULTS
##
##  MODEL 1
##
##  Model Coefficients - perf_eval
##  ──────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────────────────────────────────────────────────────────
##    Predictor     Estimate     SE            Lower         Upper        t            p             Stand.
Estimate    Lower         Upper
##  ──────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────────────────────────────────────────────────────────
##    Intercept     4.4816125    0.06627859    4.35128708    4.6119379    67.617803    < .0000001
##    interviewC    0.1641895    0.04938642    0.06707963    0.2612993     3.324588     0.0009730
0.1925996    0.07868659    0.3065126
##    gender        0.6016578    0.12305528    0.35969090    0.8436248     4.889330     0.0000015
0.2832480    0.16933502    0.3971611
##  ──────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────────────────────────────────────────────────────────
##
##
##  MODEL 2
##
##  Model Coefficients - perf_eval
##  ──────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────────────────────────────────────────────────────────
##    Predictor     Estimate      SE            Lower         Upper          t            p             Sta
nd. Estimate    Lower         Upper
##  ──────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────────────────────────────────────────────────────────
##    Intercept      4.5334239    0.06838962     4.3989463     4.66790142    66.288187    < .0000001
##    interviewC     0.2591013    0.06002020     0.1410810     0.37712171     4.316903     0.0000203
0.3039343     0.1654926     0.44237609
##    gender         0.7243668    0.12999579     0.4687503     0.97998327     5.572233    < .0000001
0.3410169     0.2206779     0.46135582
##    int_gender    -0.2837635    0.10378043    -0.4878316    -0.07969549    -2.734268     0.0065501
```

```
-0.1984969    -0.3412456    -0.05574820
##  ──────────────
```

```
##
## Call:
## lm(formula = scale(perf_eval) ~ scale(interviewC) + scale(gender) +
##     scale(interviewC * gender), data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.63205 -0.53786  0.02473  0.57231  2.76703
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 2.482e-16  4.631e-02   0.000  1.00000
## scale(interviewC)           3.039e-01  7.041e-02   4.317 2.03e-05 ***
## scale(gender)               3.410e-01  6.120e-02   5.572 4.83e-08 ***
## scale(interviewC * gender) -1.985e-01  7.260e-02  -2.734  0.00655 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8993 on 373 degrees of freedom
## Multiple R-squared:  0.1978, Adjusted R-squared:  0.1913
## F-statistic: 30.65 on 3 and 373 DF,  p-value: < 2.2e-16
```

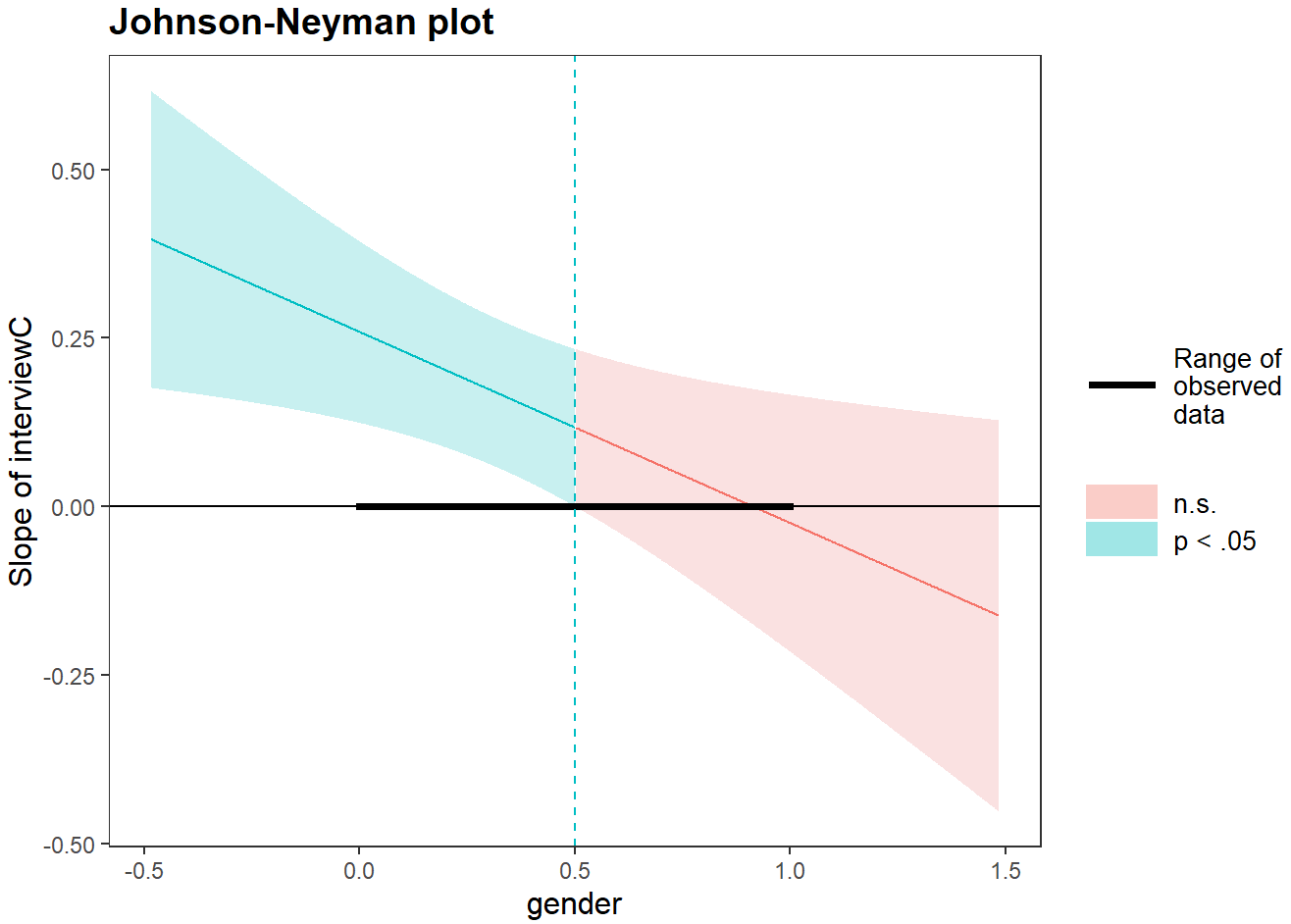
```
# summary(no_mod1Z)
```

```
# Visualization of Moderation Plots
plotSlopes(mod1,
           plotx = "interviewC",
           modx = "gender",
           main = "Gender Moderates Interview Rating Impact on Performance Eval",
           xlab = "Interview Score (Centered)",
           ylab = "Performance Evaluation",
           modxVals = "std.dev.",
           legendArgs = list(title = "Gender"))
```




```
johnson_neyman(mod1,
               pred = "interviewC",
               modx = "gender",
               control.fdr = TRUE)
```

```
## JOHNSON-NEYMAN INTERVAL
##
## When gender is OUTSIDE the interval [0.50, 3.75], the slope of interviewC
## is p < .05.
##
## Note: The range of observed values of gender is [0.00, 1.00]
##
## Interval calculated using false discovery rate adjusted t = 2.25
```



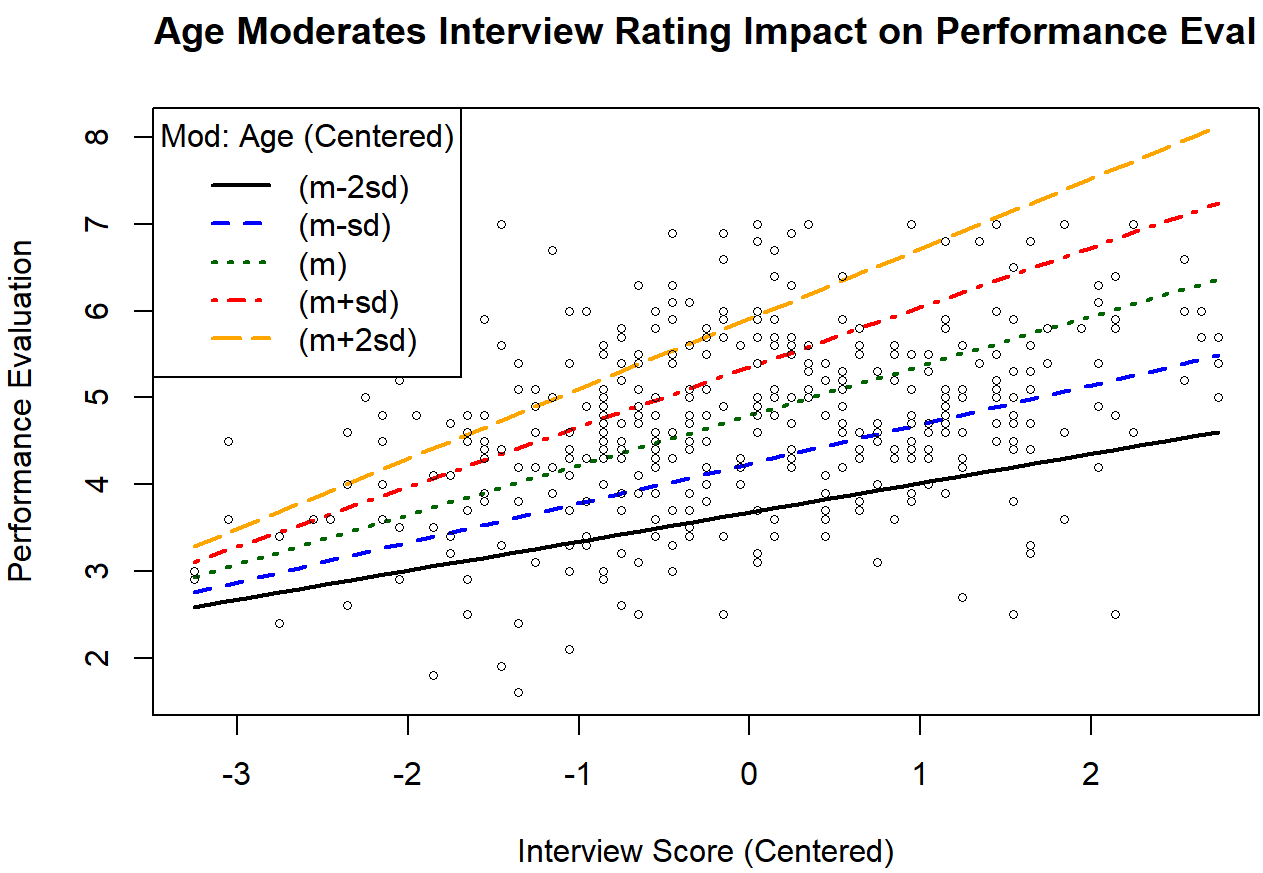
Moderation | `linReg()` Technique Interview By Age Moderation

```
linReg(data = dat,
       dep = 'perf_eval',
       covs = c('interviewC', 'ageC','int_age'),
       blocks = list(
         list('interviewC', 'ageC'),
         list('int_age')),
       ci = TRUE,
       modelTest = TRUE,
       r2Adj = TRUE,
       stdEst = TRUE,
       ciStdEst = TRUE)
```

```
##
##  LINEAR REGRESSION
##
##  Model Fit Measures
##  ─────────────────────────────────────────────────────────────────────────────────────────
##    Model    R            R²           Adjusted R²    F           df1    df2    p
##  ─────────────────────────────────────────────────────────────────────────────────────────
##        1    0.5062494    0.2562884      0.2523113    64.44156      2    374    < .0000001
##        2    0.5310845    0.2820508      0.2762764    48.84512      3    373    < .0000001
##  ─────────────────────────────────────────────────────────────────────────────────────────
##    Note. Models estimated using sample size of N=377
##
##
##  Model Comparisons
##  ─────────────────────────────────────────────────────────────────────────────────────────
##    Model         Model    ΔR²           F           df1    df2    p
##  ─────────────────────────────────────────────────────────────────────────────────────────
##        1    -        2    0.02576237    13.38446      1    373    0.0002902
##  ─────────────────────────────────────────────────────────────────────────────────────────
##
##
##  MODEL SPECIFIC RESULTS
##
##  MODEL 1
##
##  Model Coefficients - perf_eval
##  ─────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────────
##    Predictor     Estimate      SE             Lower         Upper         t             p             St
and. Estimate    Lower        Upper
##  ─────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────────
##    Intercept     4.70503979    0.045766318    4.61504823    4.79503134    102.805731    < .0000001
##    interviewC    0.55761543    0.049316718    0.46064263    0.65458824     11.306824    < .0000001
0.6541011    0.5403488    0.7678534
##    ageC          0.05914438    0.007403700    0.04458628    0.07370247      7.988489    < .0000001
0.4621351    0.3483828    0.5758873
##  ─────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────────
##
##
##  MODEL 2
##
##  Model Coefficients - perf_eval
##  ─────────────────────────────────────────────────────────────────────────────────────────
────────────────────────────────────────
##    Predictor     Estimate      SE             Lower          Upper         t            p             St
and. Estimate    Lower         Upper
##  ─────────────────────────────────────────────────────────────────────────────────────────
────────────────────────────────────────
##    Intercept     4.79527182    0.051339311    4.694321061    4.89622258    93.403510    < .0000001
##    interviewC    0.57216327    0.048682601    0.476436511    0.66789002    11.752931    < .0000001
0.6711662    0.55887559    0.7834568
##    ageC          0.06947120    0.007811884    0.054110345    0.08483205     8.893014    < .0000001
0.5428255    0.42280076    0.6628503
##    int_age       0.01467149    0.004010271    0.006785914    0.02255706     3.658478     0.0002902
```

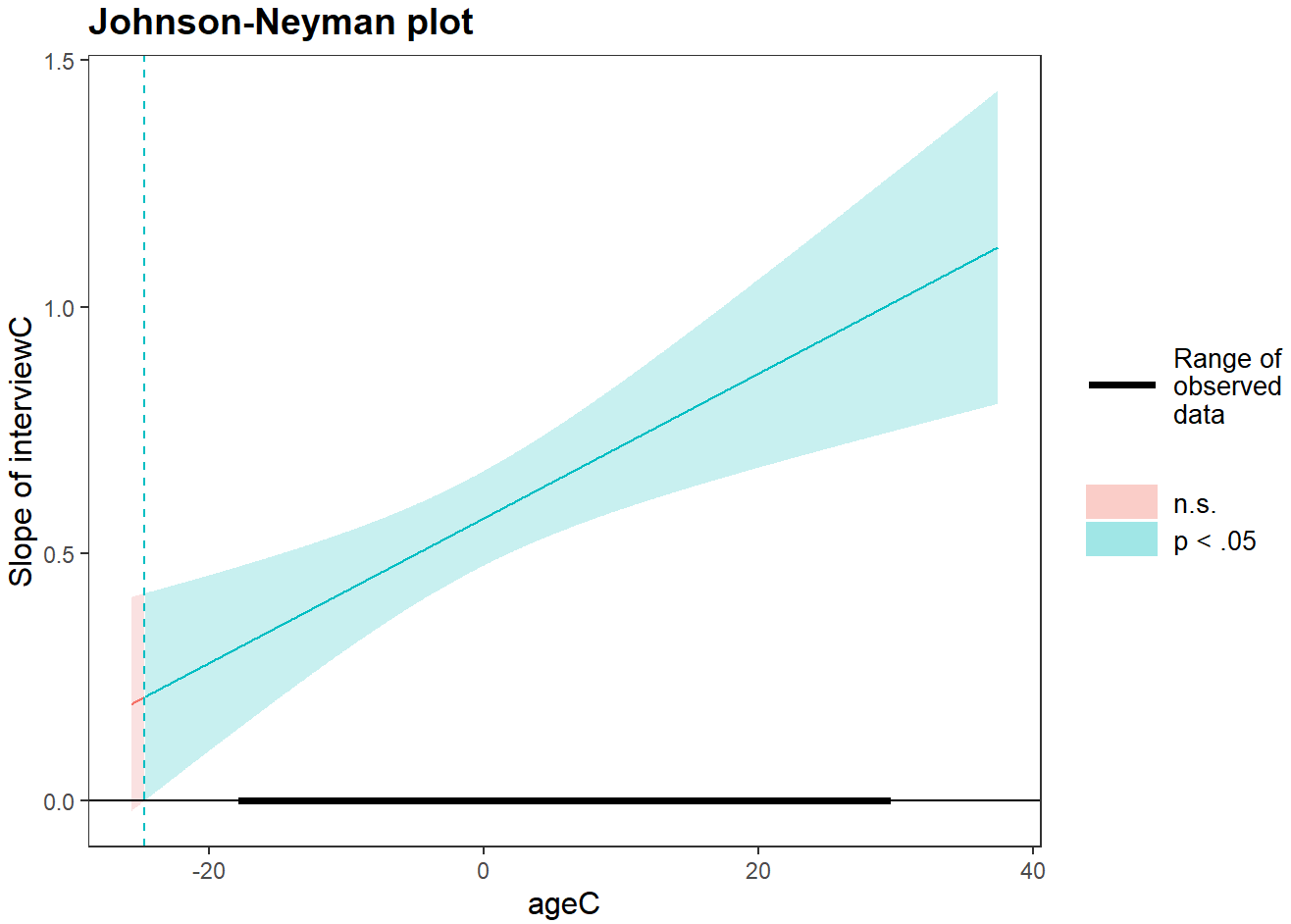
```
0.1755282    0.08118600    0.2698704
##  ──────────────
```

```
# Visualization of Moderation Plots
plotSlopes(mod2,
           plotx = "interviewC",
           modx = "ageC",
           main = "Age Moderates Interview Rating Impact on Performance Eval",
           xlab = "Interview Score (Centered)",
           ylab = "Performance Evaluation",
           modxVals = "std.dev.",
           n = 5,
           legendArgs = list(title = "Mod: Age (Centered)"))
```




```
johnson_neyman(mod2,
               pred = "interviewC",
               modx = "ageC",
               control.fdr = TRUE)
```

```
## JOHNSON-NEYMAN INTERVAL
##
## When ageC is OUTSIDE the interval [-84.18, -24.69], the slope of interviewC
## is p < .05.
##
## Note: The range of observed values of ageC is [-17.64, 29.36]
##
## Interval calculated using false discovery rate adjusted t = 1.97
```



Mediation Tutorial Markdown

We are going to use `dat2` for a tutorial on mediation analyses. This data set consists of customer product review data. A product was reviewed by customers based on how much they enjoy the product (*Enjoy*), the quality of the product (*Qual*), how excited the customer was to purchase the item (*Excite*), how happy the purchase made the customer (*Happy*), and how much intent went into the purchase (*Intent*).

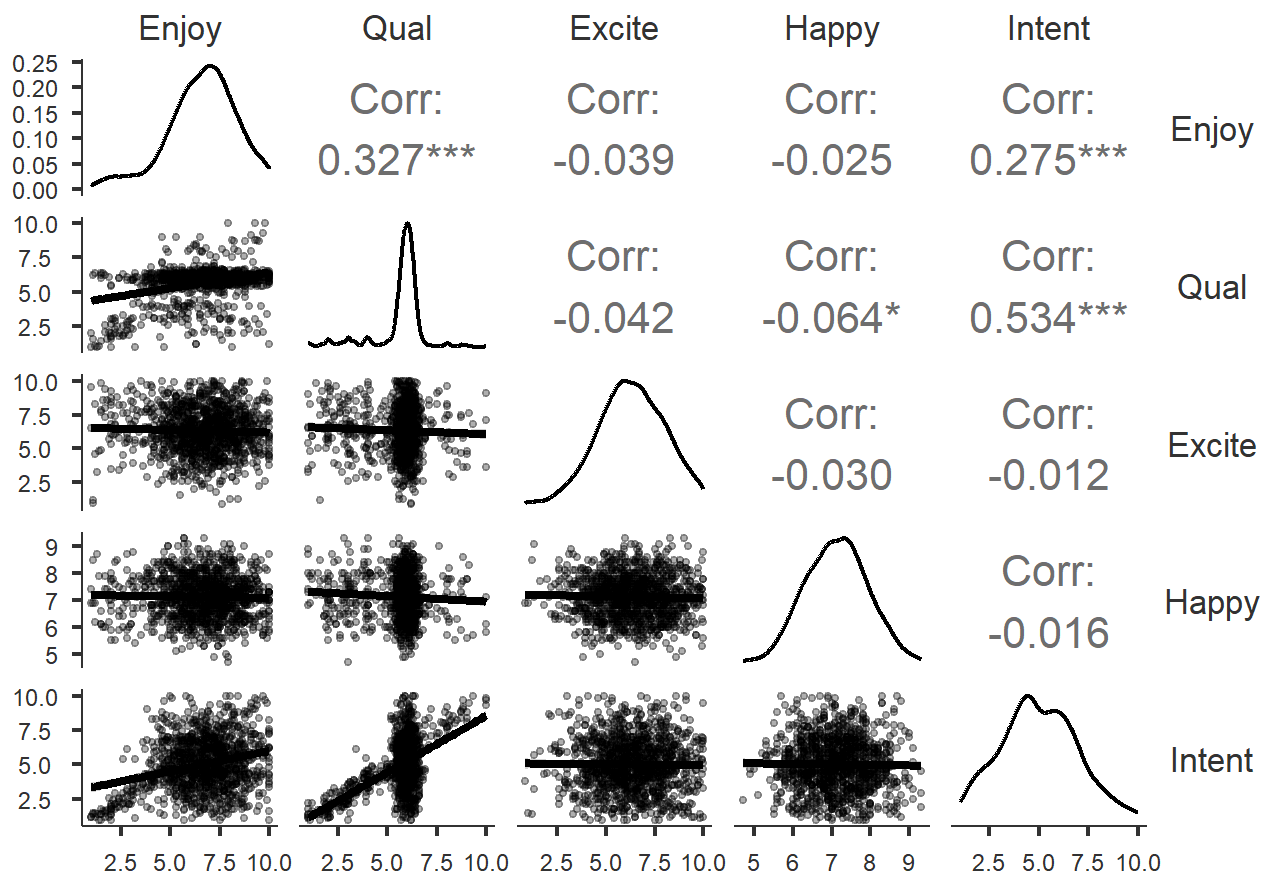
Applied prompt: The product manager wants to know how enjoyment of the product impacts quality ratings and whether the association between these variables can be accounted for by the intent of purchasing the product. We will treat quality as the criterion variable for this tutorial, with enjoyment and intent as predictor and mediator, respectively.

Calculate Descriptive Statistics and Correlation Matrix

```
describe(dat2[2:6])
```

```
##        vars    n mean   sd median trimmed  mad  min  max range  skew kurtosis
## Enjoy     1 1150 6.60 1.77    6.8    6.70 1.63 1.00 10.0  9.00 -0.59     0.45
## Qual      2 1150 5.65 1.24    5.9    5.85 0.44 1.00 10.0  9.00 -1.45     3.73
## Excite    3 1150 6.30 1.70    6.3    6.32 1.78 0.87 10.0  9.13 -0.13    -0.25
## Happy     4 1150 7.12 0.79    7.1    7.12 0.89 4.70  9.3  4.60  0.02    -0.31
## Intent    5 1150 5.00 1.91    4.9    4.97 1.93 1.00 10.0  9.00  0.15    -0.41
##          se
## Enjoy  0.05
## Qual   0.04
## Excite 0.05
## Happy  0.02
## Intent 0.06
```

```
corrMatrix(dat2[2:6],
           plots = TRUE,
           plotDens = TRUE,
           plotStats = TRUE)$plot
```



Mediation | `linReg()` Technique

```
#Indirect Path (a): M ~ X

linReg(data=dat2,
       dep=Intent,
       covs=c(Enjoy),
       blocks=list(
         list('Enjoy')),
       ci = TRUE,
       modelTest = TRUE,
       r2Adj = TRUE,
       stdEst = TRUE,
       ciStdEst = TRUE)
```

```
##
##  LINEAR REGRESSION
##
##  Model Fit Measures
##  ─────────────────────────────────────────────────────────────────────────────────────────────
##    Model    R            R²            Adjusted R²    F           df1    df2     p
##  ─────────────────────────────────────────────────────────────────────────────────────────────
##        1    0.2748678    0.07555232     0.07474705    93.82258      1    1148    < .0000001
##  ─────────────────────────────────────────────────────────────────────────────────────────────
##    Note. Models estimated using sample size of N=1150
##
##
##  MODEL SPECIFIC RESULTS
##
##  MODEL 1
##
##  Model Coefficients - Intent
##  ─────────────────────────────────────────────────────────────────────────────
```

```
# Standardized Direct Path (c)
lm1Z <- lm(scale(Qual)~scale(Enjoy), dat2)
lm1 <- lm(Qual ~ Enjoy, dat2)

# Standardized Indirect Path (a)
lm2Z <- lm(scale(Intent) ~ scale(Enjoy), dat2)
lm2 <- lm(Intent ~ Enjoy, dat2)

# Standardized Direct Path (c') and Indirect Path (b)
lm3Z <- lm(scale(Qual) ~ scale(Enjoy) + scale(Intent), dat2)
lm3 <- lm(Qual ~ Enjoy + Intent, dat2)
```

```
# Model Comparison for Mediation Effect
anova(lm1Z, lm3Z)
```

```
## Analysis of Variance Table
##
## Model 1: scale(Qual) ~ scale(Enjoy)
## Model 2: scale(Qual) ~ scale(Enjoy) + scale(Intent)
##   Res.Df     RSS Df Sum of Sq     F    Pr(>F)
## 1   1148 1026.37
## 2   1147  781.13  1    245.23 360.1 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Model Statistic Summaries
summary(lm1Z)
```

```
##
## Call:
## lm(formula = scale(Qual) ~ scale(Enjoy), data = dat2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.2213 -0.1969  0.1601  0.4667  3.2727
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -6.633e-16  2.788e-02    0.00        1
## scale(Enjoy)  3.267e-01  2.789e-02   11.71   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9455 on 1148 degrees of freedom
## Multiple R-squared:  0.1067, Adjusted R-squared:  0.106
## F-statistic: 137.2 on 1 and 1148 DF,  p-value: < 2.2e-16
```

```
summary(lm2Z)
```

```
##
## Call:
## lm(formula = scale(Intent) ~ scale(Enjoy), data = dat2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.62268 -0.69007 -0.02905  0.63435  2.86714
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -7.178e-16  2.836e-02   0.000        1
## scale(Enjoy)  2.749e-01  2.838e-02   9.686   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9619 on 1148 degrees of freedom
## Multiple R-squared:  0.07555,    Adjusted R-squared:  0.07475
## F-statistic: 93.82 on 1 and 1148 DF,  p-value: < 2.2e-16
```

```
summary(lm3Z)
```

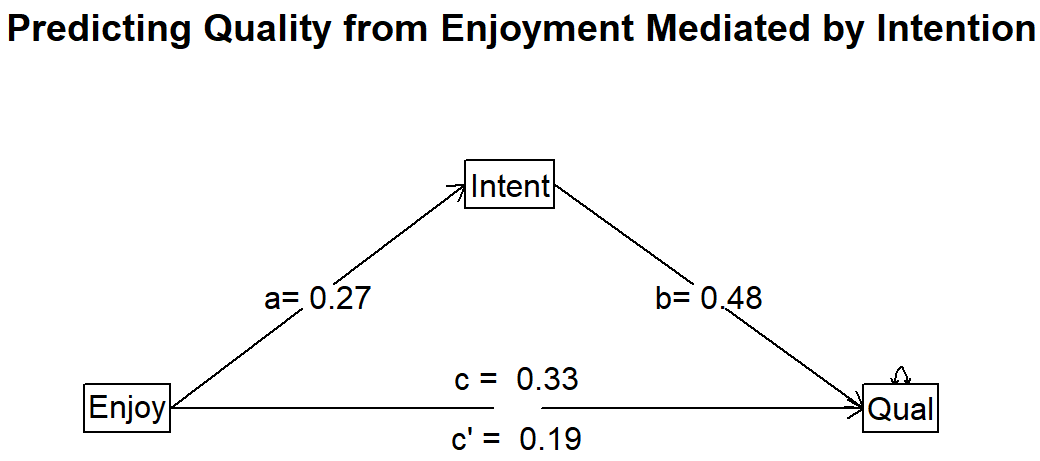
```
##
## Call:
## lm(formula = scale(Qual) ~ scale(Enjoy) + scale(Intent), data = dat2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.2158 -0.3949  0.0633  0.5528  2.2875
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -3.184e-16  2.434e-02   0.000        1
## scale(Enjoy)   1.946e-01  2.532e-02   7.686 3.24e-14 ***
## scale(Intent)  4.805e-01  2.532e-02  18.976  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8252 on 1147 degrees of freedom
## Multiple R-squared:  0.3202, Adjusted R-squared:  0.319
## F-statistic: 270.1 on 2 and 1147 DF,  p-value: < 2.2e-16
```

```
# Conducting Sobel's Indirect Effect Test
sobel(pred=dat2$Enjoy,
      med=dat2$Intent,
      out=dat2$Qual)
```

```
## $`Mod1: Y~X`
##              Estimate Std. Error  t value      Pr(>|t|)
## (Intercept) 4.1429833 0.13326961 31.08723 1.944383e-154
## pred        0.2284179 0.01950322 11.71181  5.230990e-30
##
## $`Mod2: Y~X+M`
##              Estimate Std. Error   t value      Pr(>|t|)
## (Intercept) 3.1943037 0.12660257 25.230955 4.319906e-112
## pred        0.1360759 0.01770375  7.686275  3.240738e-14
## med         0.3116312 0.01642218 18.976240  4.756102e-70
##
## $`Mod3: M~X`
##              Estimate Std. Error   t value     Pr(>|t|)
## (Intercept) 3.0442381 0.20904014 14.562936 3.267375e-44
## pred        0.2963183 0.03059178  9.686206 2.211602e-21
##
## $Indirect.Effect
## [1] 0.09234202
##
## $SE
## [1] 0.01070349
##
## $z.value
## [1] 8.627283
##
## $N
## [1] 1150
```

Visualization of Mediation Model

```
mediate(Qual ~ Enjoy + (Intent),
        data = dat2,
        std = TRUE,
        main = "Predicting Quality from Enjoyment Mediated by Intention"
        )$plot
```


Predicting Quality from Enjoyment Mediated by Intention
Intent
a= 0.27
b= 0.48
c = 0.33
Enjoy
Qual
c' = 0.19

```
## NULL
```